MULTIMEDIA UNIVERSITY

CDS 6324

DATA VISUALIZATION

PROJECT (40%)

Lecture Section: TC2L

Tutorial Section: TT2L

Group Number: TT2L_G12

Submitted to: Dr. Noramiza Binti Hashim

Student ID	Student Name	Email & Contact No
1221305612	Yaser E H Abulaban	1221305612@student.mmu.edu.my +6017-7039024
1221305898	Chan Ga Wai	1221305898@student.mmu.edu.my +6011-64129980
1221305145	Mishal Mann Nair	1221305145@student.mmu.edu.my +6019-4432285

Contents

Proposal Preparation Work Distribution

Student Name	**Student ID**	Proposal Preparation Responsibilities
Yaser E H Abulaban	1221305612	• Prepared the dataset description • Checked dataset suitability against the project requirements • Reviewed key dataset attributes • Contributed to the analytical questions.
Chan Ga Wai	1221305898	• Designed the initial dashboard storyboard/layout • Proposed the spatial and ranking visualizations • Prepared the explanation for the choropleth map and ranked country bar chart.
Mishal Mann Nair	1221305145	• Prepared the project description • Refined the interaction plan • Contributed to the time-based and composition visualizations • Reviewed the proposal formatting and consistency.
Group Contribution	All Members	• Discussed the dashboard concept • Selected the SDG 3 focus area • Reviewed the proposed visualizations • Finalized the proposal content together.

Project Implementation Work Distribution

Student Name	**Student ID**	Main Focus Area
Yaser E H Abulaban	1221305612	• Temporal STI trend analysis • Multi-Line Time-Series Chart • Animated Bubble Chart • Dashboard integration
Chan Ga Wai	1221305898	• Spatial and country ranking analysis • Choropleth World Map • Ranked Country Bar Chart • Global filters
Mishal Mann Nair	1221305145	• Year-over-year change and disease composition analysis • Year-over-Year Lollipop Chart • Disease Share Donut Chart • Linked interactions
Group Contribution	All Members	• Testing • Report writing • Presentation preparation

Project Title

Global STI Trends Dashboard: Interactive Monitoring of HIV, Gonorrhea, and Syphilis Burden

Assigned SDG Theme

SDG 3: Good Health & Well-being

Proposed Dataset(s) and Attributes

Global STI Intelligence: HIV·Gonorrhea ·Syphilis

https://www.kaggle.com/datasets/kanchana1990/global-sti-intelligence-hivgonorrhea-syphilis

The proposed dataset contains global sexually transmitted infection indicators related to HIV, gonorrhea, and syphilis. The dataset consists of 67438 records and 31 attributes, covering data from 1990 to 2024 across 200 countries. It includes disease-related information, country and regional information, time-based attributes, numerical health indicator values, burden classifications, and year-over-year trend indicators.

The temporal dimension is represented by attributes such as year, decade, is_recent, and years_ago, while the spatial dimension is represented by attributes such as country_code, country_alpha2, country_name, and who_region. The dataset is relevant to SDG 3 because sexually transmitted infections are important global public-health issues.

Column Name	Description
record_id	Unique identifier for each record
disease	Name of the STI, such as HIV, Gonorrhea, or Syphilis
disease_type	Disease category, such as Viral or Bacterial
disease_treatment	Main treatment approach, such as ART, antibiotics, or penicillin
is_curable	Indicates whether the disease is curable; 1 = curable, 0 = not curable
indicator_code	Code representing the health indicator
metric	Specific public-health measurement or indicator
country_code	Three-letter country code
country_alpha2	Two-letter country code
country_name	Name of the country
who_region	WHO region, such as Africa, Americas, or Europe
year	Year of the observation
decade	Decade grouping, such as 1990s or 2000s
is_recent	Indicates whether the record is considered recent
years_ago	Number of years before the dataset scrape date
sex	Original sex or subgroup code used by the source
sex_label	Display label for sex group; in this dataset it is mainly Both Sexes
value	Main numerical value for the selected metric
value_low	Lower estimate or confidence interval value

value_high	Upper estimate or confidence interval value
value_display	Formatted version of the value for display
ci_width	Width of confidence interval
burden_tier	Burden classification such as LOW, MEDIUM, HIGH, or VERY HIGH
value_prev_year	Value from the previous year
yoy_change_abs	Absolute year-over-year change
yoy_change_pct	Percentage year-over-year change
yoy_direction	Direction of change, such as INCREASED, DECREASED, or STABLE
data_source	Original data source name
source_url	URL of the source API or dataset
scrape_date	Date when the dataset was collected
dataset_version	Version of the dataset

Because the dataset contains multiple public-health indicators, the dashboard will include a **Metric / Indicator filter**. This is important because the value column may represent different measurements depending on the selected metric. To avoid misleading comparisons, all charts will display results based on the same selected metric at one time. The data will also be cleaned and prepared before visualization by removing unnecessary fields, checking missing values, standardizing country and year fields and aggregating records where needed by country, year, disease and selected metric.

Project Description

The proposed dashboard will be designed as a single-page interactive web dashboard titled "Global STI Trends Dashboard". The dashboard will use the Global STI Intelligence: HIV, Gonorrhea, and Syphilis dataset to help users explore sexually transmitted infection indicators across countries, years, diseases, disease types, and burden classifications. The dashboard focuses on public-health monitoring, STI burden, and trend analysis.

The dashboard will be structured to guide users from a broad global overview to more detailed country-level and trend-based analysis. Users will first view summary indicators and a global map, then explore country rankings, time trends, year-over-year changes, and disease composition through linked interactive visualizations.

Analytical Questions

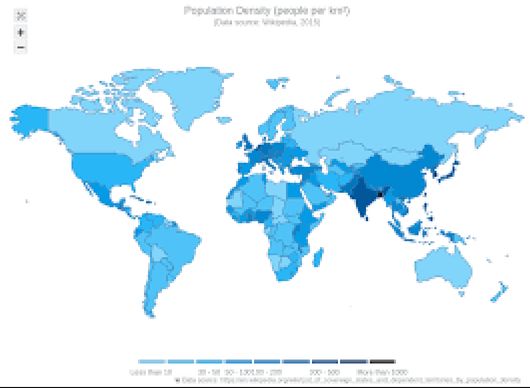
This dashboard is designed to answer the following analytical questions:

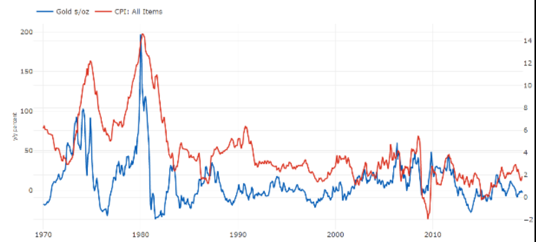
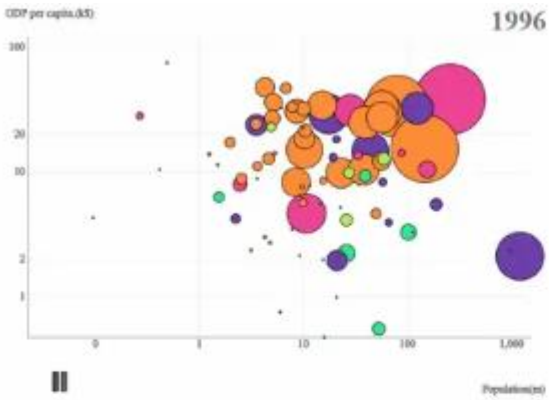
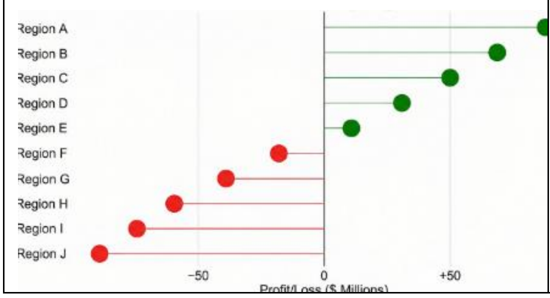
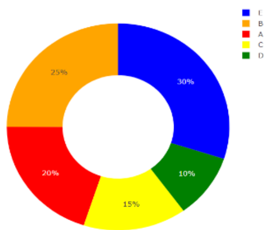
1. **Which countries and WHO regions have the highest STI burden for HIV, gonorrhea, and syphilis in a selected year?**
   This question will be answered using the choropleth world map, ranked country bar chart and KPI cards.
2. **How have STI indicators changed from 1990 to 2024 across countries and diseases?**
   This question will be answered using the multi-line time-series chart and animated bubble chart.
3. **Which countries show the fastest worsening or improvement in STI indicators based on year-over-year change?**
   This question will be answered using the year-over-year lollipop chart and the fastest worsening/improving KPI cards.
4. **Which STI contributes the largest share of the selected health indicator under different filters?**
   This question will be answered using the disease share donut chart and linked dashboard filters.

These questions guide the visualization choices and help the dashboard move from a general data display to a focused public-health analysis tool.

Dashboard Interface Layout

The dashboard interface layout below is a draft and may not be the final dashboard layout. The data shown in the charts during the proposal stage is for planning purposes only and does not represent the final cleaned dataset.

<table>
<tr><th colspan="7">Global STI Trends Dashboard</th></tr>
<tr><td>Filter by</td><td>Metric / Indicator</td><td>Disease</td><td>Disease Type</td><td>Year / Year Range</td><td>Country / WHO Region</td><td>Burden Tier</td></tr>
<tr><td colspan="7">KPI Cards</td></tr>
<tr><td>Global Burden Scores</td><td>Highest Burden Country</td><td>Lowest Burden Country</td><td>Dominant Disease</td><td>Fastest Worsening Country</td><td colspan="2">Fastest Improving Country</td></tr>
<tr><td colspan="3">Choropleth World Map
</td><td colspan="4">Ranked Country Bar Chart

</td></tr>
<tr><td colspan="3">Multi-Line Time-Series Chart

</td><td colspan="4">Animated Bubble Chart

</td></tr>
<tr><td colspan="3">Year-over-Year Lollipop Chart

</td><td colspan="4">Disease Share Donut Chart
</td></tr>
</table>

At the top of the dashboard, the title will clearly indicate the purpose of the system. Below the title, a filter panel will allow users to customize the displayed data. The available filters will include Metric / Indicator, Disease, Disease Type, Year or Year Range, Country, WHO Region and Burden Tier.

The Metric / Indicator filter will allow users to select the specific public-health measurement to analyze, ensuring that the dashboard does not mix different indicator types. The Disease filter will allow users to select HIV, gonorrhea, or syphilis. The Disease Type filter will allow users to compare broader disease categories such as viral and bacterial STIs. The Year or Year Range filter will support time-based exploration from 1990 to 2024. The Country and WHO Region filters will allow users to examine global, regional, or country-specific patterns. The Burden Tier filter will help users focus on records classified as low, medium, high, or very high burden.

Below the filter section, the dashboard will display KPI cards that summarize the selected data context. The KPI cards will include Global Burden Score, Highest Burden Country, Lowest Burden Country, Dominant Disease, Fastest Worsening Country, and Fastest Improving Country. These indicators provide users with an immediate overview of the global STI situation before they explore the detailed charts. The Global Burden Score summarizes the selected indicator value across available countries. The Highest and Lowest Burden Country cards identify extreme values. The Dominant Disease card shows which disease contributes the largest share of the selected metric, while the Fastest Worsening and Fastest Improving Country cards highlight recent changes based on year-over-year values.

The table below illustrates the KPI cards planned and its definitions.

KPI Card	Definition
Global Burden Score	The aggregated value of the selected metric based on the current filters. The dashboard will display the selected metric name to avoid confusion.
Highest Burden Country	The country with the highest value for the selected metric, disease, and year.
Lowest Burden Country	The country with the lowest value for the selected metric, disease, and year.
Dominant Disease	The disease with the largest share of the selected metric under the current filters.
Fastest Worsening Country	The country with the highest positive year-over-year percentage change for the selected metric.
Fastest Improving Country	The country with the largest negative year-over-year percentage change for the selected metric.

The KPI cards provide users with an immediate summary before they explore the detailed visualizations. They also support decision-making by highlighting countries or diseases that may require closer public-health attention.

The table below explains the visualization planned and its justification.

Visualization	Intended Purpose	Justification
Choropleth World Map	Shows STI burden by country	A map is suitable because the dataset contains spatial attributes such as country_name. It helps users identify geographical patterns and high-burden areas.
Ranked Country Bar Chart	Compares countries based on the selected metric	A bar chart is suitable for ranking and precise comparison. It complements the map by showing exact country comparisons more clearly.
Multi-Line Time-Series Chart	Shows how STI indicators change over time	A line chart is suitable because the dataset contains yearly data from 1990 to 2024. It supports the analytical goal of identifying long-term trends.
Animated Bubble Chart	Shows country-level STI burden changes across years	Animation is suitable because the dataset has temporal data. Bubble size can represent indicator value, while movement across years shows progression over time.
Year-over-Year Lollipop Chart	Highlights countries with the largest increase or decrease	A lollipop chart is suitable for showing ranked change values clearly. It helps users identify countries where STI indicators are worsening or improving most.
Disease Share Donut Chart	Shows which disease contributes the largest share of the selected metric	A donut chart is suitable because there are only three disease categories and supports part-to-whole comparison.

The dashboard will include several interactive features to support data exploration. Hover tooltips will display details such as country, WHO region, disease, year, selected metric, value, burden tier and year-over-year change. Filtering will allow users to narrow the dataset by metric, disease, disease type, year, country, WHO region and burden tier.

Zooming and panning will be implemented in the choropleth world map so users can explore smaller countries and dense regions more easily. The multi-line time-series chart may include brush zooming to help users focus on a specific period.

The dashboard will also include linked interactions. Selecting a country on the choropleth map or ranked bar chart will update the time-series chart, animated bubble chart, and year-over-year lollipop chart. Selecting a disease segment in the donut chart will update the other charts to focus on that disease. These linked interactions allow users to move from a global overview to detailed country-level and disease-level analysis without manually adjusting every chart.

The animated bubble chart will include play, pause, and year-slider controls. This animation will show how STI burden changes across countries over time. The animation is analytical rather than decorative because it helps users understand temporal progression in global STI indicators.